

AMENDMENT NO. 3 JUNE 2018
TO
IS 3597 : 1998 CONCRETE PIPES — METHODS OF TEST

(Second Revision)

[(Page 4, clause 7.2.4 (see also Amendment No. 1)] — Substitute the following for the existing clause:

‘7.2.4 Pressure shall be maintained by continuous pumping the water using throttling device for 1 min + 30 s for each 10 mm of wall thickness (for precast concrete pipes wall thickness shall be full barrel wall thickness, whereas it shall be core thickness, in case of prestressed concrete pipe) or for twice that entire period, if the application of pressure resulted in the formation of beads of water on the pipe surface.’

(Page 5, Fig. 5) — Insert the following new clause at the end:

‘10 SOCKET AND SPIGOT DIAMETERS

10.1 Socket Diameter (Jointing Surface)

10.1.1 Procedure

The socket diameter shall be checked by measuring the sway by touching the two points B and B_1 along the circumference of socket (see Fig. 6) BB_1 gives the sway.

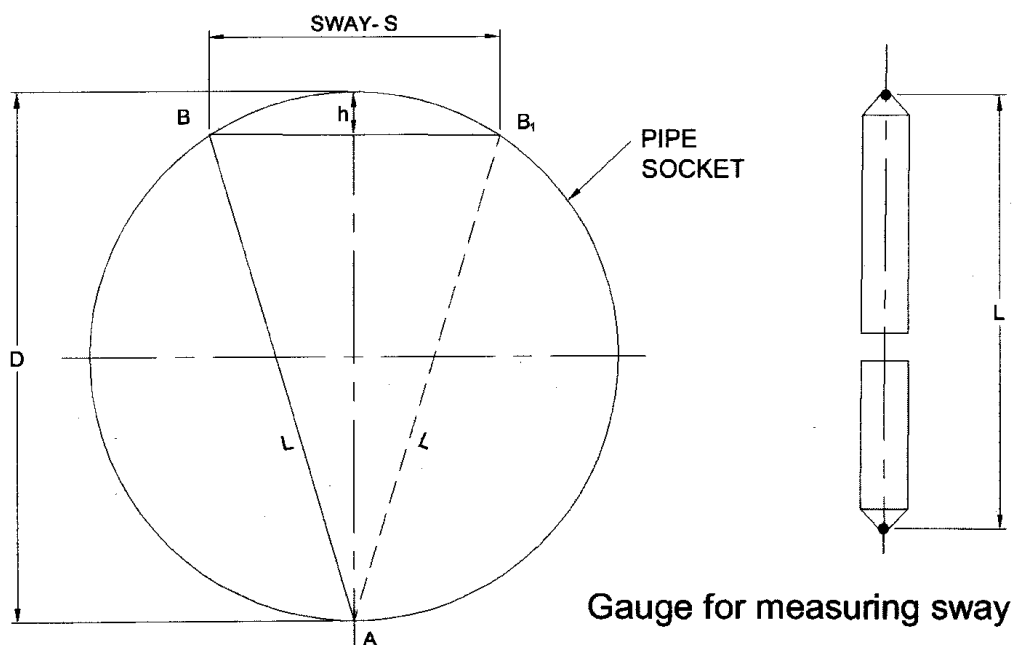


FIG. 6 ARRANGEMENT FOR MEASURING SWAY

Example:

Suppose the theoretical diameter of socket, D = 1 383 mm

Permissible variation = -0.5^{+2} mm

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The maximum diameter, D_1	= 1 385 mm
The minimum diameter, D_2	= 1 382.5 mm
The sway, S	= $2 \times [h \times (D_1 - h)]^{1/2}$
Assume h (see Fig. 6)	= 5 mm
Sway, S	= $2 \times [5 \times (1\,385 - 5)]^{1/2}$
	= $2 \times [5 \times (1\,385 - 5)]^{1/2}$
	= 2×83
	= 166 mm
Length of gauge, L	
L^2	= $(S/2)^2 + (D_1 - h)^2$
	= $(166/2)^2 + (1\,385 - 5)^2$
	= 6 889 + 1 904 400
L	= 1 382.5 mm
This should not be more than the minimum permissible inside diameter.	
Therefore, in this case, L	= 1 382.5 mm
Maximum permissible sway	= 166 mm

10.2 Spigot Diameter

10.2.1 Procedure

The gauge as shown in Fig. 7 shall be used.

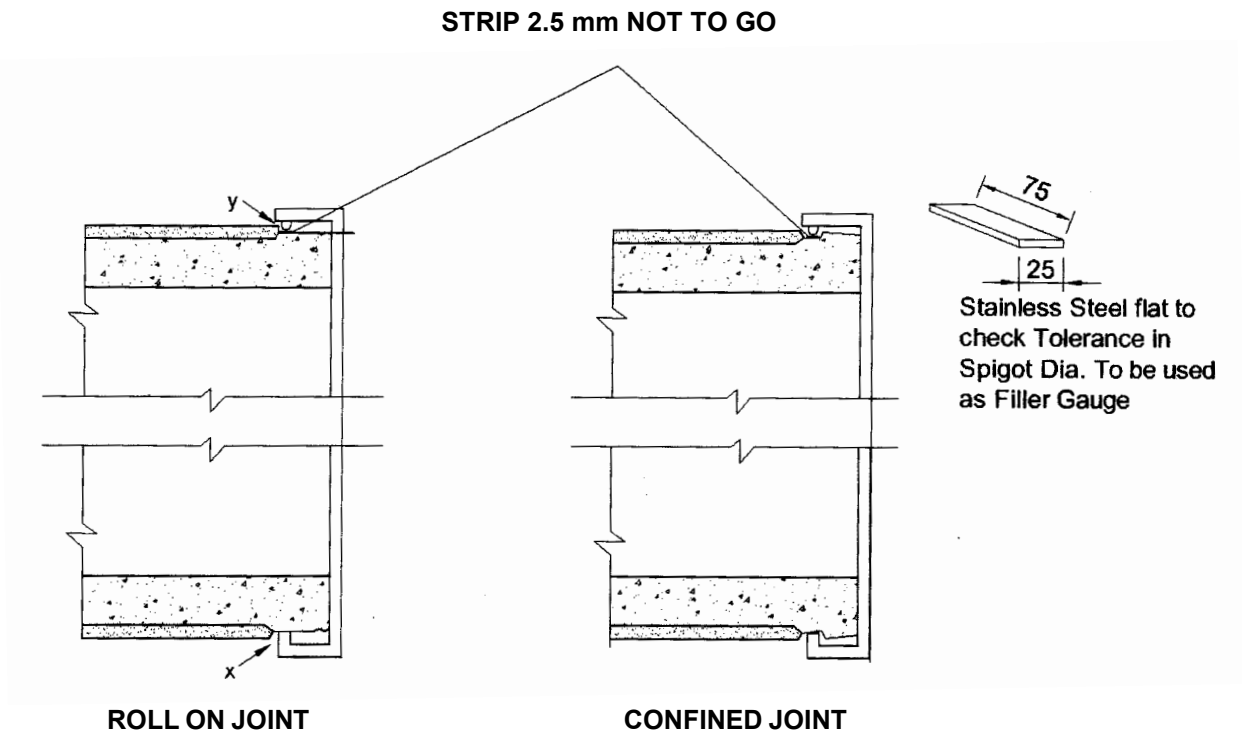


FIG. 7 ARRANGEMENT FOR CHECKING SPIGOT DIAMETER

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Example:

Suppose the theoretical outside diameter of spigot	= 1 200 mm
Permissible variation	= ± 1 mm
The maximum diameter	= 1 201 mm
The minimum diameter	= 1 199 mm

To enable gauge to pass over the spigot surface, the diameter must be more by 1mm than the maximum spigot diameter.

$$\begin{aligned}\text{The clearance between gauge points} &= 1\,200 + 1 + 1 \\ &= 1\,202\text{ mm}\end{aligned}$$

For checking spigot, the gauge is held in the position as shown in Fig. 7. Then by keeping point 'X' fixed, other end 'Y' of gauge is moved over the circumference of spigot, where the gap between spigot surface and gauge is maximum, a strip 2.5 mm thick and 25 mm wide (straight side) is inserted. The available gap being $1\,201 - 1\,199 = 2$ mm, it should not go. Two checks shall be done at 90° to each other.

11 BOILING ABSORPTION TEST

11.1 Test Specimen

Core pipe / lined cylinder pipe shall be fixed on coating machine and soap solution shall be applied on outside surface of plain pipe barrel for about 1 m in length. The cement mortar coating having thickness 22 to 25 mm shall be applied on the pipe at this portion. Coated pipe shall be removed from coating machine and shall be placed on sand bed. After 10 to 15 min, 25 to 30 mm width grooves shall be made on green coating along the longitudinal direction up to the full depth of cement mortar coating at 150 mm spacing. Green coating shall be covered with reinforced plastic sheet immediately. Coated pieces shall be taken out on the next day and shall be cured in the same manner as pipe is cured. Pieces shall be cut to get samples of size 100×100 mm by using concrete cutter.

11.2 Test Procedure

Three specimens of uniform coating thickness shall be selected for test and shall be put in oven at a temperature of 105 to 115°C for minimum 24 h. Oven-dried specimens shall be weighed immediately after removal from the oven. Specimens shall be allowed to cool to the room temperature. After cooling of specimens, it shall be placed in tap/distilled water having temperature in the range of 10 - 24°C. Water shall be heated to boil and the boiling should take place in not less than 1 h and not more than 2 h. Boiling of the water shall be continued for further 5 h. At the end of 5 h boiling period, heat shall be turned off and samples shall be allowed to cool in the same water to room temperature by **natural water cooling period of 14h to 24h**. Specimens shall be removed from the water and shall be allowed to drain by placing on an open drain rack for 1 min. Remaining superficial water shall be removed by blotting the specimen with a dry absorbent/blotting paper. Specimens shall be weighed immediately after blotting.

11.3 Calculations

Increase in mass of boiled specimen over its dry mass shall be calculated as the water absorption by specimen and shall be expressed it as a percentage of the dry mass.

Percentage for each specimen shall be calculated as below:

Mass of dry specimen (Oven dried) = M_1

Mass of specimen after 5 h boiling and **natural water cooling period of 14h to 24h** = M_2

$$\text{Boiling water absorption of specimen in percentage} = \frac{M_2 - M_1}{M_1} \times 100$$

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12 PRESSURE PERMEABILITY TEST ON CEMENT MORTAR COATING OF FINISHED PIPES

12.1 Procedure for Pressure Permeability Test

12.1.1 Apparatus

The apparatuses shown in Fig. 8, shall consist of,

- a) permeability cup of dimensions of 3 000 mm² inside cross-sectional area with clamping arrangement on the coating of the finished pipe.
- b) rubber pad for water tight sealing between permeability cup and pipe coating surface.
- c) graduated acrylic tube attached to permeability cup with a suitable pressure gauge at its upper end.
- d) a source of compressed air (compressor/foot rested pump) to give a regulated constant pressure of 3 ± 0.2 kg/cm² connected at the upper end of acrylic tube by hose pipe.
- e) mild steel clamping device and eye bolts with wire ropes for clamping cup on pipe coating surface.

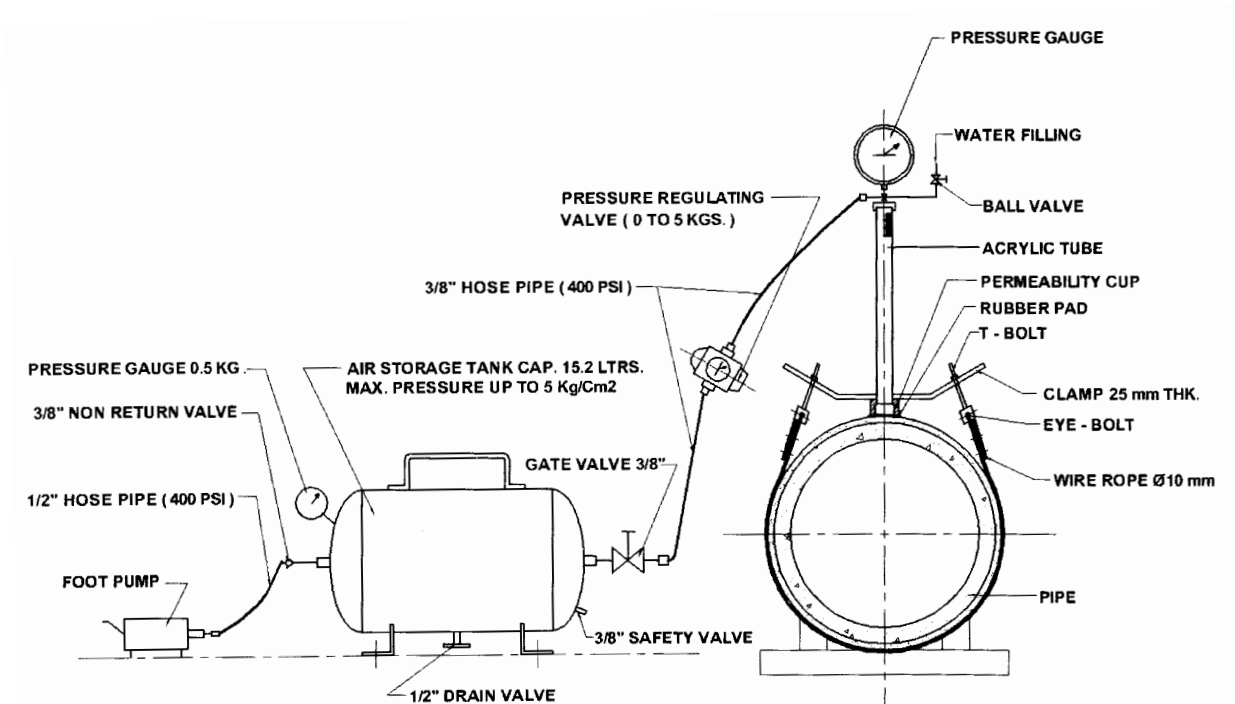


FIG. 8 TYPICAL PRESSURE PERMEABILITY TEST ARRANGEMENT

12.1.2 Procedure

- a) The cover coat of the pipe shall be fully saturated with water prior to testing.
- b) Scrap the dry coating of the pipe by wire brush, remove loose particles, if any.
- c) Clamp the equipment on coating as shown in Fig. 8 and fill the water in acrylic tube up to level by wash bottle.

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- d) Install the pressure gauge at top of the tube using Tee connection and compressed air source to be fitted to Tee connection.
- e) Chock whole assembly shall be checked for leaks under the required pressure of 3 kg/cm². If leak is observed, tight the fittings to get water tight connection.
- f) Maintain the pressure of 3 kg/cm² by using pressure regulator throughout the test.
- g) Record the drop in water levels at 60 min intervals under sustain pressure of 3 kg/cm² up to 6 h.

12.1.3 Test Acceptance Criteria

The ratio of cumulative absorption per hour to area of cup orifice in contact with the pipe surface is to be calculated. The ratio shall not exceed 0.15 / h for the period between 121 and 180 min. If this requirement has not been attained in the above stipulated period, the test is to be continued for further periods either 240, 300, 360, 420 or 480 min until the requirement is met. If the requirement has not been met at 480 min, the pipe has failed to the test, and in such case further 2 pipes from the batch shall be tested. If both pipes pass the test the batch shall be accepted, otherwise the same shall be rejected.